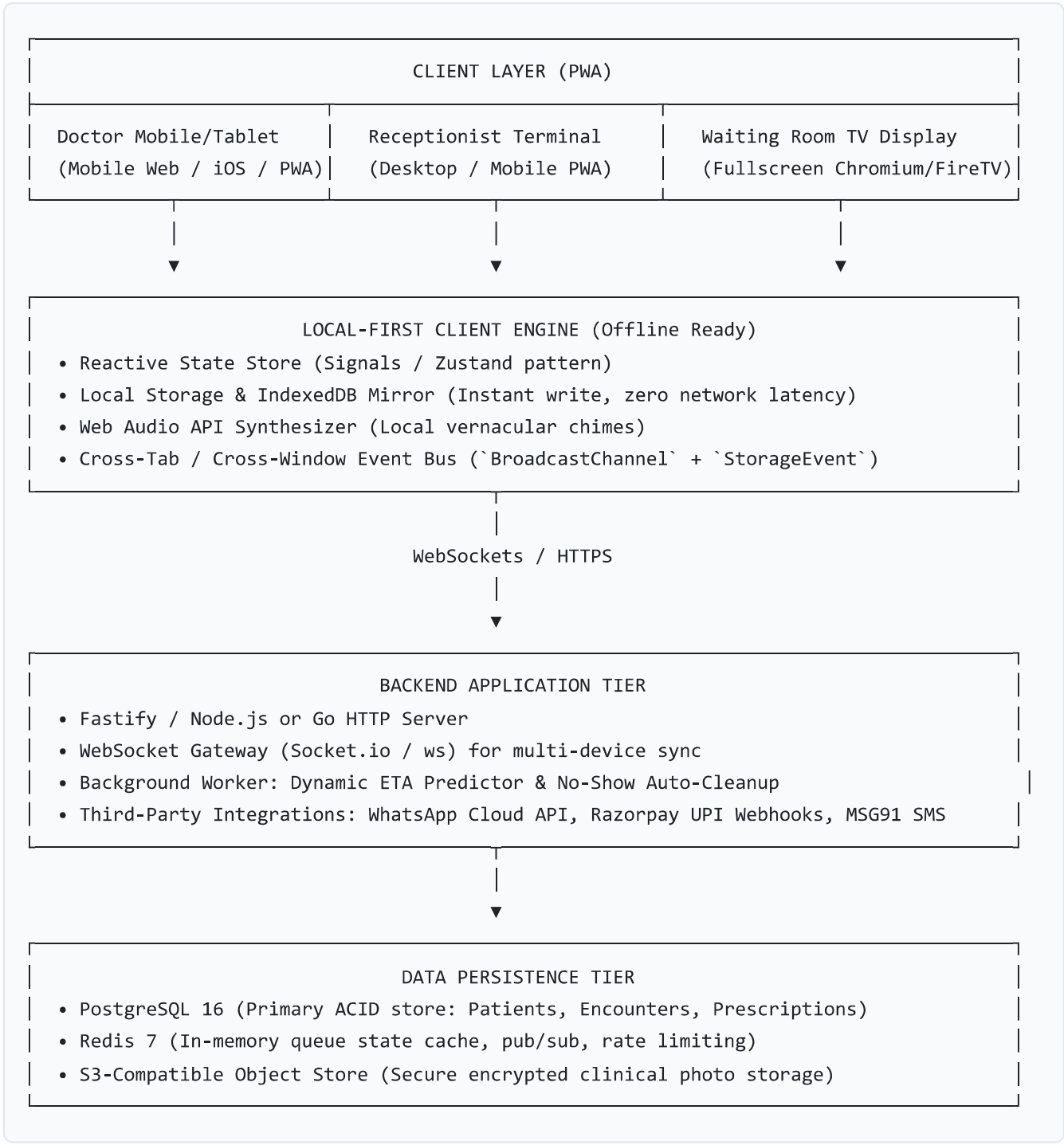
📐 Document 2: Technical Design Document (TDD)

Product Name: ClinikFlow (India Specialist Edition)
Document Version: 1.0.0
Target Architecture: Mobile-First Progressive Web Application (PWA) with Edge Synchronization
Status: Approved for Engineering Implementation

1. System Architecture Overview

ClinikFlow is designed as a **hybrid local-first, cloud-synchronized** clinical queue and EMR system. Because broadband reliability in Indian Tier-2/Tier-3 clinic locations can fluctuate, the application maintains state locally via `IndexedDB` and `BroadcastChannel` for same-network displays, while syncing changes bidirectionally with a central cloud API when online.


CLIENT LAYER (PWA)
Doctor Mobile/Tablet
(Mobile Web / iOS / PWA)
Receptionist Terminal
(Desktop / Mobile PWA)
Waiting Room TV Display
(Fullscreen Chromium/FireTV)
LOCAL-FIRST CLIENT ENGINE (Offline Ready)
• Reactive State Store (Signals / Zustand pattern)
• Local Storage & IndexedDB Mirror (Instant write, zero network latency)
• Web Audio API Synthesizer (Local vernacular chimes)
• Cross-Tab / Cross-Window Event Bus (`BroadcastChannel` + `StorageEvent`)
WebSockets / HTTPS
BACKEND APPLICATION TIER
• Fastify / Node.js or Go HTTP Server
• WebSocket Gateway (Socket.io / ws) for multi-device sync
• Background Worker: Dynamic ETA Predictor & No-Show Auto-Cleanup
• Third-Party Integrations: WhatsApp Cloud API, Razorpay UPI Webhooks, MSG91 SMS
DATA PERSISTENCE TIER
• PostgreSQL 16 (Primary ACID store: Patients, Encounters, Prescriptions)
• Redis 7 (In-memory queue state cache, pub/sub, rate limiting)
• S3-Compatible Object Store (Secure encrypted clinical photo storage)

2. Database Schema (PostgreSQL DDL)

```
sql
-- Extensions
CREATE EXTENSION IF NOT EXISTS "uuid-ossp";
CREATE EXTENSION IF NOT EXISTS "pgcrypto";

-- 1. Clinics Master
CREATE TABLE clinics (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    clinic_name VARCHAR(255) NOT NULL,
    slug VARCHAR(100) UNIQUE NOT NULL,
    doctor_name VARCHAR(255) NOT NULL,
    specialty VARCHAR(100) NOT NULL, -- 'PEDIATRICS', 'DERMATOLOGY', 'ORTHOPEDICS', etc.
    phone VARCHAR(15) NOT NULL,
    address TEXT,
    upi_vpa VARCHAR(100), -- Dynamic UPI QR target (e.g. clinic@okaxis)
    default_consultation_fee NUMERIC(8,2) DEFAULT 500.00,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);

-- 2. Patients Directory (Phone-Centric Identity)
CREATE TABLE patients (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    clinic_id UUID NOT NULL REFERENCES clinics(id) ON DELETE CASCADE,
    uhid VARCHAR(30) NOT NULL, -- Starlight format: ST-YYMM-XXXXXX
    phone VARCHAR(15) NOT NULL,
    full_name VARCHAR(255) NOT NULL,
    age INT NOT NULL,
    age_unit VARCHAR(10) DEFAULT 'YEARS', -- 'YEARS', 'MONTHS', 'DAYS'
    gender VARCHAR(10) NOT NULL, -- 'MALE', 'FEMALE', 'OTHER'
    blood_group VARCHAR(5),
    allergies TEXT[] DEFAULT ARRAY[]::TEXT[],
    chronic_conditions TEXT[] DEFAULT ARRAY[]::TEXT[],
    created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),
    updated_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),
    CONSTRAINT uq_clinic_phone UNIQUE(clinic_id, phone),
    CONSTRAINT uq_clinic_uhid UNIQUE(clinic_id, uhid)
);
CREATE INDEX idx_patients_phone ON patients(phone);
CREATE INDEX idx_patients_uhid ON patients(uhid);

-- 3. Queue Tokens (Daily Ephemeral & Audit State)
CREATE TYPE token_status AS ENUM (
    'WAITING',
```

```
    'CALLED',
    'IN_CONSULTATION',
    'HOLD_FOR_LAB',
    'REPORT_REVIEW',
    'COMPLETED',
    'NO_SHOW',
    'CANCELLED'
);

CREATE TYPE priority_tier AS ENUM (
    'NORMAL',
    'EMERGENCY',
    'SENIOR_CITIZEN',
    'REPORT_REVIEW'
);

CREATE TABLE queue_tokens (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    clinic_id UUID NOT NULL REFERENCES clinics(id) ON DELETE CASCADE,
    patient_id UUID NOT NULL REFERENCES patients(id) ON DELETE RESTRICT,
    token_number INT NOT NULL,
    queue_date DATE NOT NULL DEFAULT CURRENT_DATE,
    status token_status NOT NULL DEFAULT 'WAITING',
    priority priority_tier NOT NULL DEFAULT 'NORMAL',
    chief_complaint TEXT,
    fee_amount NUMERIC(8,2) NOT NULL DEFAULT 500.00,
    payment_status VARCHAR(20) NOT NULL DEFAULT 'PENDING', -- 'PAID_CASH', 'PAID_UPI',
'PENDING'
    upi_txn_ref VARCHAR(100),
    estimated_consult_time TIMESTAMP WITH TIME ZONE,
    called_at TIMESTAMP WITH TIME ZONE,
    started_at TIMESTAMP WITH TIME ZONE,
    completed_at TIMESTAMP WITH TIME ZONE,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),
    CONSTRAINT uq_daily_token UNIQUE(clinic_id, queue_date, token_number)
);
CREATE INDEX idx_queue_active ON queue_tokens(clinic_id, queue_date, status);

-- 4. Clinical Encounters (EMR Lite Records)
CREATE TABLE clinical_encounters (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    token_id UUID NOT NULL REFERENCES queue_tokens(id),
    patient_id UUID NOT NULL REFERENCES patients(id),
    clinic_id UUID NOT NULL REFERENCES clinics(id),
    vitals JSONB NOT NULL DEFAULT '{}'::JSONB,
```

```
    -- vitals format: {"bp_systolic": 120, "bp_diastolic": 80, "pulse": 72, "weight_kg":
65, "temp_f": 98.4, "spo2": 99, "rbs": 110}
    specialist_data JSONB NOT NULL DEFAULT '{}'::JSONB,
    -- structured data tailored per specialty
    provisional_diagnosis TEXT[] DEFAULT ARRAY[]::TEXT[],
    rx_items JSONB NOT NULL DEFAULT '[]'::JSONB,
    -- rx format: [{"brand": "Augmentin 625", "generic": "Amoxiclav", "dosage": "1-0-1",
"timing": "After Food", "days": 5}]
    advice_notes TEXT,
    follow_up_date DATE,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);
```

3. Real-Time Synchronization & Local-First Strategy

To support both **single-machine offline clinics** (laptop connected via HDMI to TV) and **networked multi-device clinics** (doctor on iPad, receptionist on PC, Android TV on wall), ClinikFlow implements a dual-tier synchronization protocol:

Tier 1: Local Bus Synchronization (`BroadcastChannel`)

For devices running on the same browser/client profile or local LAN:

- Channel: `clinik_local_sync`
- Event Types:
  - `TOKEN_ISSUED`: Broadcasted by Receptionist → → TV and Doctor Pad update immediately.
  - `TOKEN_CALLED`: Broadcasted by Doctor → → TV plays vernacular audio and animates "Now Serving".
  - `TOKEN_COMPLETED`: Moves token to completed list and clears waiting room count.

Tier 2: Cloud WebSocket Relay (`Socket.io`)

When connected to the internet:

- Rooms: `clinic:{clinic_id}`
- The backend broadcasts state changes down to connected clients within 50ms.
- Patients on their personal mobile devices (`clinik.in/q/{token_id}`) receive passive updates without polling.

4. Dynamic Wait-Time Prediction Engine

Standard linear wait-time models fail in Indian OPDs because doctor consultation duration fluctuates based on patient rush.

The Algorithm:

$$\text{WaitTime}_{\text{token } k} = \sum_{i=1}^{k-1} (\hat{\tau}_{\text{consult}} \times \lambda_{\text{rush}}) - \Delta t_{\text{elapsed}}$$

WaitTimetoken k = i=1∑k−1 (τ^consult × λrush) − Δtelapsed

Where:

1. $\hat{\tau}_{\text{consult}}$ τ^consult: **Exponential Moving Average (EMA)** of the last 5 completed consultations today: $\hat{\tau}_t = \alpha \cdot \text{ActualDuration}_t + (1-\alpha) \cdot \hat{\tau}_{t-1} \quad (\alpha = 0.35)$τ^t = α · ActualDurationt + (1 − α) · τ^t−1 (α = 0.35) *(Defaults to 6.5 minutes if fewer than 3 patients have been seen).*
2. λ_{rush} λrush: **Queue Pressure Damping Factor**:
   - If waiting queue > 15> 15 patients: $\lambda_{\text{rush}} = 0.85$λrush = 0.85 (doctors naturally accelerate).
   - If waiting queue ≤ 5≤ 5 patients: $\lambda_{\text{rush}} = 1.05$λrush = 1.05 (doctors spend more time explaining).
3. $\Delta t_{\text{elapsed}}$ Δtelapsed: Time elapsed in the current patient's consultation.

5. WhatsApp & Thermal Print Integration Specs

1. WhatsApp Cloud API Webhook Dispatch

- **Trigger:** Token creation at receptionist desk.
- **Payload:**

```json
{
  "messaging_product": "whatsapp",
  "to": "919876543210",
  "type": "template",
  "template": {
    "name": "clinic_token_confirmation_v2",
    "language": { "code": "hi" },
    "components": [
      {
        "type": "body",
        "parameters": [
          { "type": "text", "text": "Sunita Devi" },
          { "type": "text", "text": "18" },
          { "type": "text", "text": "Dr. Rajesh Iyer" },
          { "type": "text", "text": "14" },
          { "type": "text", "text": "22 mins" },
          { "type": "text", "text": "https://clinik.in/q/7f8a9b1c" }
        ]
      }
    ]
  }
}
```

2. Thermal Slip Generator (Web Bluetooth / Raw ESC/POS)

- **Dimensions:** 58mm (2-inch) / 80mm (3-inch).
- **Buffer Layout:**

```text
[ESC @]               -- Initialize printer
[ESC a 1]             -- Align Center
** DR. IYER'S CLINIC **
Token No: [GS ! 11] #18 [GS ! 00]
Patient: Sunita Devi (42F)
Time: 14-Oct-2026 06:30 PM
Now Serving: #14
--------------------------------
[GS k 4] clinik.in/q/7f8a9b1c  -- QR Code
Scan to track queue on WhatsApp
[ESC d 3]             -- Feed 3 lines
[GS V 66 0]           -- Paper cut
```

6. Security, Privacy & Compliance (DISHA & ABDM)

1. **Patient Identifier Masking:** The public TV screen never displays full phone numbers or sensitive diagnostic complaints. It displays only `Token Number` and `First Name` (e.g., `#18 - Sunita D.`).
2. **Access Control:**
    - Compounder role: Can view queue, demographics, and payments. Cannot view clinical diagnosis or consultation notes.
    - Doctor role: Unrestricted access to medical records, prescriptions, and financials.
3. **Audit Logging:** Every status change (`CALLED`, `HOLD`, `COMPLETED`) is immutably timestamped with user ID.